

52 cards distributed evenly among

4 players

(a) one of the players receives all 13 Hearts

E - event that one of the players receives all 13 Hearts

E_i - event that i^{th} player receives all 13 Hearts.

$$P(E_i) = \frac{1}{\binom{52}{13}}, \quad i = 1, 2, 3, 4$$

$$P(E) = P(E_1) + P(E_2) + P(E_3) + P(E_4).$$

$$= \frac{1}{\binom{52}{13}} + \frac{1}{\binom{52}{13}} + \frac{1}{\binom{52}{13}} + \frac{1}{\binom{52}{13}}$$

$$= \frac{4}{\frac{52!}{13! 39!}}$$

$$= \frac{4 (13!) 39!}{52!}$$

(b) each player receives 1 ace.

F - event that each player receives 1 ace

There are only 4 ace in a pack of 52 cards

First distribute 4 ace among 4 players evenly

$$\begin{array}{cccc}
 \square & \square & \square & \square \\
 \uparrow & \uparrow & \uparrow & \uparrow \\
 4 & 3 & 2 & 1
 \end{array}
 \quad (\text{OR}) \quad
 \begin{pmatrix} 4 \\ 1 & 1 & 1 & 1 \end{pmatrix}$$

4! ways for distributing ace among 4 players evenly.

We are left with 48 cards, distribute them evenly among 4 players

$$\begin{array}{cccc}
 \square & \square & \square & \square \\
 \uparrow & \uparrow & \uparrow & \uparrow \\
 \binom{48}{12} & \times & \binom{36}{12} & \times & \binom{24}{12} & \times & \binom{12}{12}
 \end{array}
 \quad (\text{OR}) \quad
 \begin{pmatrix} 48 \\ 12 & 12 & 12 & 12 \end{pmatrix}$$

$$\Rightarrow \frac{48!}{(12!)^4}$$

The desired probability

$$P(F) = 4! \frac{48!}{(12!)^4}$$

$$= 0.1055.$$

Problem ③

Let n people are present in a room.

① What is the probability that no two of them celebrate birthday on the same day of the year?

② How big the number ' n ' must be so that the probability is less than $\frac{1}{2}$?

Solution :

Sample space S consists of all possible outcomes of birthdays

of n people in the given year
(ans the year is not a leap year)

$\square \square \square \dots \square$ n people -

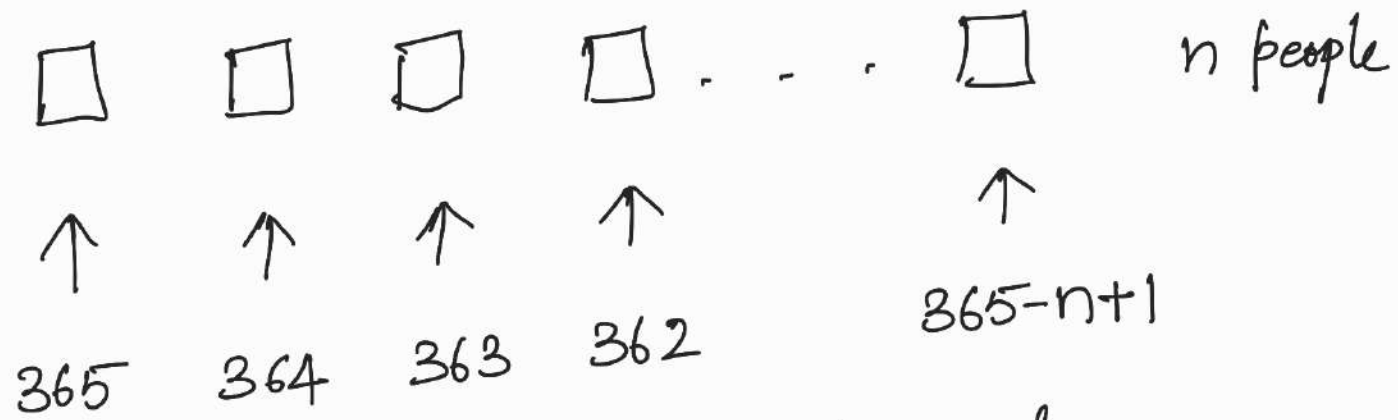
$\uparrow \quad \uparrow \quad \uparrow \quad \dots \quad \uparrow$
 $365 \quad 365 \quad 365 \quad \dots \quad 365$

Each person can have birthday on any
one of 365 days of the year.

Basic principle of counting,

$$n(s) = 365 \times 365 \times \dots \times 365 \quad (n \text{ times})$$
$$= (365)^n.$$

E - event that no two of these n
people celebrate their birthday on
the same day.



Applying basic principle of counting

$$n(E) = 365 \times 364 \times 363 \times \dots \times (365-n+1)$$

$$P(E) = \frac{n(E)}{n(S)}$$

$$P(E) = \frac{365 \times 364 \times \dots \times (365-n+1)}{(365)^n}$$

② what should be n so that

$$P(E) < \frac{1}{2}$$

$$\text{put } n=22 \longrightarrow P(E) \geq \frac{1}{2}$$

$$\text{put } n=23 \longrightarrow P(E) < \frac{1}{2}$$

$$n \geq 23$$

$$P(E) < \frac{1}{2}$$

In a room there must be at least 23 people, so that the probability of no two of them having same birth day is $< 1/2$.

problem ④

A total of 36 members of a club play tennis, 28 play squash and 18 play badminton.

Further, 22 members play both tennis and squash, 12 play both tennis and badminton, 9 play both squash and badminton, 4 play all sports.

How many members of this club play at least one of three sports?

Solution: Let N denote the number of members of the club.

T - set of members play tennis
 S - set of members play squash
 B - set of members play Badminton

$$p(T) = \frac{36}{N} ; p(S) = \frac{28}{N}$$

$$p(B) = \frac{18}{N}$$

$$p(T \cap S) = \frac{22}{N} ; p(T \cap B) = \frac{12}{N}$$

$$p(S \cap B) = \frac{9}{N} ; p(T \cap B \cap S) = \frac{4}{N}$$

$$p(\overset{F}{T \cup \underbrace{S \cup B}_{F}})$$

$$= p(T) + \underbrace{p(S \cup B)}_{\downarrow} - \underbrace{p(T \cap (S \cup B))}_{\downarrow}$$

$$= P(T) + P(S) + P(B) - P(S \cap B) \\ - P((T \cap S) \cup (T \cap B))$$

$$= P(T) + P(S) + P(B) - P(S \cap B) \\ - \left[P(T \cap S) + P(T \cap B) - P(T \cap S \cap B) \right]$$

$$= P(T) + P(S) + P(B) \\ - P(S \cap B) - P(T \cap S) - P(T \cap B) \\ + P(T \cap S \cap B)$$

$$= \frac{36}{N} + \frac{28}{N} + \frac{18}{N} - \frac{9}{N} - \frac{22}{N} \\ - \frac{12}{N} + \frac{4}{N}$$

$$= \frac{43}{N}$$

Therefore, 43 members of the club play atleast one of the sports.

EX: Suppose $E_1, E_2, E_3, \dots, E_N$
are events of an experiment

then

$$P\left(\bigcup_{i=1}^N E_i\right)$$

$$= \sum_{i=1}^N P(E_i) - \sum_{i_1 < i_2} P(E_{i_1} \cap E_{i_2})$$

$$+ \dots + (-1)^{n+1} \sum_{i_1 < i_2 < \dots < i_n} P(E_{i_1} \cap E_{i_2} \cap \dots \cap E_{i_n})$$

$$+ \dots + (-1)^{N+1} P(E_1 \cap \dots \cap E_N)$$

The matching problem

Suppose that each of N men at a party throws his hat into the center of the room.

First hats are mixed up and then each man randomly selects a hat.

What is the probability that none of the men selects his own hat?

Solution:

E_i - event that the i^{th} man
select his own hat

$$i=1, 2, 3, \dots, N$$

The probability that none of
them selects his own hat is

$$P(E_1^c \cap E_2^c \cap E_3^c \cap \dots \cap E_N^c)$$

$$= P\left(\left(\bigcup_{i=1}^N E_i\right)^c\right)$$

$$= \underline{\underline{1 - P\left(\bigcup_{i=1}^N E_i\right)}}$$

$$P\left(\bigcup_{i=1}^N E_i^c\right)$$

$$= \sum_{i=1}^N P(E_i) - \sum_{i_1 < i_2} P(E_{i_1} \cap E_{i_2}) \\ + \dots + (-1)^{r+1} \sum_{i_1 < i_2 < \dots < i_r} P(E_{i_1} \cap \dots \cap E_{i_r}) \\ + \dots + (-1)^{N+1} P(E_1 \cap \dots \cap E_N).$$

Firstly, we compute

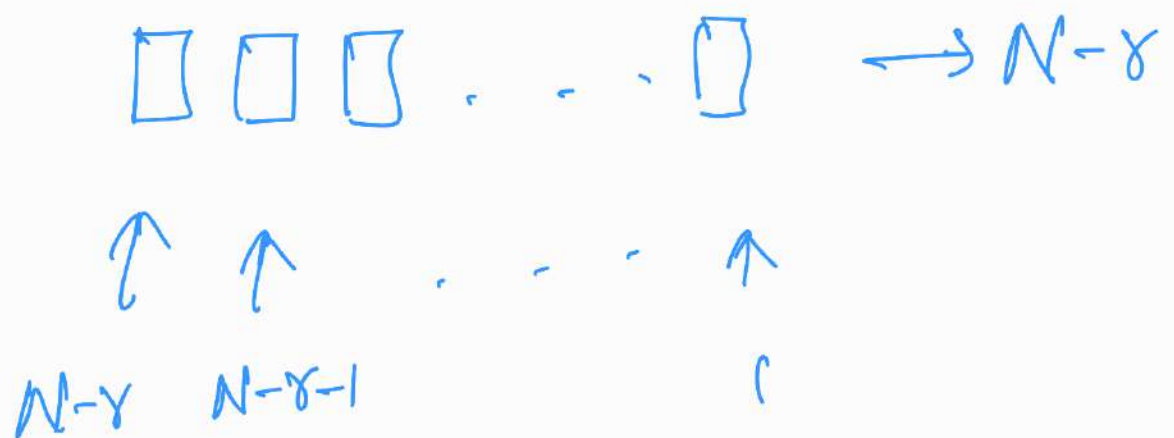
$$P(E_{i_1} \cap E_{i_2} \cap \dots \cap E_{i_r}).$$

There are $N!$ ways for N men
selecting hats.

Here the event $E_{i_1} \cap E_{i_2} \cap \dots \cap E_{i_r}$

denotes that r men selecting their own hat.

The rest of the $(N-r)$ men select a hat from the remaining $(N-r)$ hats



there are $(N-r)!$ ways

in which r men select their own hat

$$P(E_{i_1} \cap E_{i_2} \cap \dots \cap E_{i_r})$$

$$= \frac{(N-r)!}{N!}$$

In the summation

$$\sum_{i_1 < i_2 < \dots < i_r} P(E_{i_1} \cap E_{i_2} \cap \dots \cap E_{i_r})$$

there are $\binom{N}{r}$ terms

$$\therefore \sum_{i_1 < i_2 < \dots < i_r} P(E_{i_1} \cap E_{i_2} \cap \dots \cap E_{i_r})$$

$$= \binom{N}{r} \frac{(N-r)!}{N!_0}$$

$$= \frac{\cancel{N!}_0}{(\cancel{N-r})!_0 r!} \frac{(\cancel{N-r})!_0}{\cancel{N!}_0}$$

$$= \frac{1}{r!_0}$$

$$\therefore P\left(\bigcup_{i=1}^N E_i^c\right)$$

$$\begin{aligned}
&= \sum_{i=1}^N P(E_i) - \dots \\
&\quad - (-1)^{r+1} \sum_{i_1 < i_2 < \dots < i_r} P(E_{i_1} \cap E_{i_2} \cap \dots \cap E_{i_r}) \\
&\quad \dots + (-1)^{N+1} P(E_1 \cap \dots \cap E_N)
\end{aligned}$$

$$\begin{aligned}
&\Rightarrow 1 - \frac{1}{2!} + \frac{1}{3!} - \dots + (-1)^{r+1} \frac{1}{r!} + \dots \\
&\quad \dots + (-1)^{N+1} \frac{1}{N!} \\
&\quad \quad \quad \underline{\underline{=}}
\end{aligned}$$

The desired probability

$$P(E_1^c \cap E_2^c \cap E_3^c \cap \dots \cap E_N^c)$$

$$\Rightarrow 1 - P\left(\bigcup_{i=1}^N E_i^c\right)$$

$$\Rightarrow 1 - \left[1 - \frac{1}{2!} + \frac{1}{3!} + \dots + (-1)^{N+1} \frac{1}{N!}\right]$$

$$\Rightarrow 1 - 1 + \frac{1}{2!} - \frac{1}{3!} + \frac{1}{4!} + \dots + (-1)^N \frac{1}{N!}$$

$$= \sum_{i=0}^N \frac{(-1)^i}{i!}$$



Problem: Compute the probability that 10 married couple are seated at random at a round table such that no wife sits next to her husband.

Proof-

n - people can be arranged around a round table $(n-1)!$ ways

Firstly, note that there are $19!$ ways of arranging

20 people (=10 couple) around
a round table.

E_i - event that the i^{th} couple
sits next to each other.

$$(i = 1, 2, 3, \dots, 10)$$

The desired probability

$$P(E_1^c \cap E_2^c \cap \dots \cap E_{10}^c)$$

$$= 1 - P\left(\bigcup_{i=1}^{10} E_i\right)$$

Now compute $P\left(\bigcup_{i=1}^{10} E_i\right)$

Ex:

Conditional probability

Suppose E and F are two events of an experiment

"Conditional probability that E occurs given that F occurred" is denoted by

$P(E/F)$ - probability of E given F .

If $P(F) > 0$, then

$$P(E/F) = \frac{P(E \cap F)}{P(F)}$$

problem: Suppose two dice are tossed. What is the probability that the sum of the numbers on two dice is 8 given that the number on the first die is 3?

solution: E - event that sum is 8

F - event that the first die shows number 3

$$E = \{ (2,6), (3,5), (4,4), (5,3), (6,2) \}$$

$$F = \{ (3,1), (3,2), (3,3), (3,4), (3,5), (3,6) \}$$

$$E \cap F = \{ (3,5) \}$$

$$P(E \cap F) = \frac{1}{36}$$

$$P(F) = \frac{6}{36} > 0$$

$$P(E/F) = \frac{P(E \cap F)}{P(F)}$$

↑
Conditional
probability

$$= \frac{\frac{1}{36}}{\frac{6}{36}}$$

$$= \frac{1}{6}$$

Problem: A coin flipped twice.
What is the conditional probability
that both flips lands on heads
given that the first flip is
head?

Solution:

$$S = \{ (H, H), (H, T), (T, H), (T, T) \}$$

B - event that both flips
lands on head

F - event that the first flip
lands on head.

$$B = \{ (H, H) \}$$

$$F = \{ (H, H), (H, T) \}$$

$$B \cap F = \{ (H, H) \}$$

$$P(B \cap F) = \frac{1}{4}$$

$$P(F) = \frac{2}{4} = \frac{1}{2} > 0$$

The Conditional probability

$$P(B|F) = \frac{P(B \cap F)}{P(F)}$$

$$= \frac{\frac{1}{4}}{\frac{1}{2}}$$
$$= \frac{1}{2}$$

Problem: In a card game bridge, the 52 cards are dealt out equally to 4 players. These players are East, West, North and South. If North and South have a total of 8 spades among them, what is the probability that East has 3 of the remaining 5 spades?

Solution:

Reduced Sample space is obtained by considering

that 26 cards are already distributed among North and South. From the remaining 26 cards, 13 cards are given to East

$$n(S) = \binom{26}{13}$$

E - event that East has 3 spades of the remaining 5 spades

$$n(E) = \binom{5}{3} \binom{21}{10}$$

$$p(E) = \frac{\binom{5}{3} \binom{21}{10}}{\binom{26}{13}}$$

$$= \frac{\frac{5!}{2! 3!} \cdot \frac{21!}{11! 10!}}{\frac{26!}{13! 13!}}$$

$$p(E) = \underline{\underline{0.33}}$$

We know that

$$P(E/F) = \frac{P(E \cap F)}{P(F)}$$

By multiplying, with $P(F)$ on both sides, we get

$$P(E \cap F) = P(F) P(E/F)$$

Problem: A student "charce" is undecided as to whether to take a French course (or) a Chemistry Course. She estimates that her probability of receiving an A grade would be $1/2$ in French Course

and $\frac{2}{3}$ in chemistry course.

If Charu decides to base her decision on the flip of a fair coin, what is the probability that she gets an A in chemistry?

Solution:

Let C be the event that Charu takes chemistry.

A be the event that she receives grade "A" in whatever

Course she takes.

The desired probability is

$$p(C \cap A)$$

$$= p(C) p(A|C)$$

$$= \left(\frac{1}{7}\right) \left(\frac{2}{3}\right)$$

$$\boxed{p(C \cap A) = \frac{1}{3}}$$



Suppose E_1, E_2, E_3 are events of an experiment.

$$P(E_1) P(E_2/E_1) P(E_3/E_1 \cap E_2)$$

$$= \frac{P(E_1) P(E_2 \cap E_1)}{P(E_1)} \frac{P(E_3 \cap E_1 \cap E_2)}{P(E_1 \cap E_2)}$$

$$= P(E_1 \cap E_2 \cap E_3)$$

$$P(E_1 \cap E_2 \cap E_3)$$

$$= P(E_1) P(E_2/E_1) P(E_3/E_1 \cap E_2)$$

The multiplication rule

$$P(E_1 \cap E_2 \cap \dots \cap E_n)$$

$$= P(E_1) P(E_2/E_1) P(E_3/E_1 \cap E_2) \dots P(E_n/E_1 \cap E_2 \cap \dots \cap E_{n-1})$$

Ex:

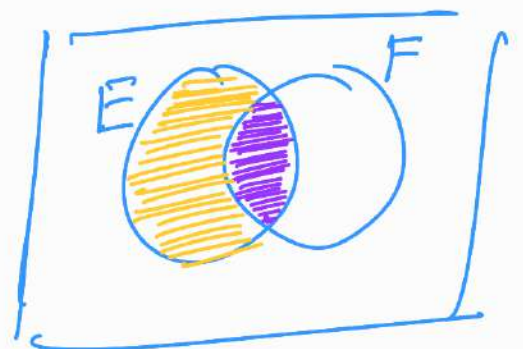


An ordinary deck of 52 playing Cards is randomly divided into 4 piles of 13 Cards each. Compute the probability that each pile has exactly 1 ace.

Bayes's Formula:

Let E and F be events

$$E = (E \cap F) \cup (E \cap F^c)$$



$$P(E) = P(E \cap F) + P(E \cap F^c)$$

$$P(E) = P(F) P(E/F) + P(F^c) P(E/F^c)$$

Problem: An insurance company believes that people can be divided into two classes:

those who are accident prone and those who are not.

The Company's statistics show that an accident prone person

will have an accident at some time within a fixed 1-year period with probability 0.4, whereas the probability decreases to 0.2 for a person in non-accident prone.

If 30% of the population is accident prone, what is the probability that a new policyholder will have an accident within a year of purchasing policy?

Solution:-

Let A_1 be the event that

the policy holder will have an accident within a year of purchasing policy.

Let A be the event that the policy holder is accident prone.

By Bayes's formula

$$p(A_1) = p(A) p(A_1/A) + p(A^c) p(A_1/A^c)$$

$$= (0.3)(0.4)$$

$$+ (0.7)(0.2)$$

$$= 0.12 + 0.14$$

$$= \underline{\underline{0.26.}}$$

$$\begin{aligned}
 & 0.45 \times 0.4 \\
 & + 0.55 \times 0.2 \\
 & = \frac{4.5}{10} \times \frac{4}{10} + \frac{5.5}{10} \times \frac{2}{10} \\
 & = 0.18 + 0.11 \\
 & = \underline{\underline{0.29}}
 \end{aligned}$$

